Solutions — Mock Olympiad (IMC), Round 2, July 19, 2026

Problem 1. (IMC 2015, Day 2, Problem 6)

We claim that for every $n \geq 1$

$$\frac{1}{\sqrt{n}\,(n+1)} \;<\; \frac{2}{\sqrt{n}} - \frac{2}{\sqrt{n+1}}. \qquad (*)$$

Multiplying $(*)$ by $\sqrt{n}\,(n+1) > 0$ and using $\frac{\sqrt{n}(n+1)}{\sqrt{n+1}} = \sqrt{n}\sqrt{n+1} = \sqrt{n(n+1)}$, inequality $(*)$ becomes

$$1 < 2(n+1) - 2\sqrt{n(n+1)}, \qquad \text{i.e.} \qquad 2\sqrt{n(n+1)} < n + (n+1),$$

which is the AM–GM inequality applied to n and $n+1$ (they are distinct, so it is strict). Summing $(*)$ over $n \geq 1$, the right-hand side telescopes:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}\,(n+1)} \;<\; \sum_{n=1}^{\infty} \left(\frac{2}{\sqrt{n}} - \frac{2}{\sqrt{n+1}} \right) = 2. \qquad \blacksquare$$

Problem 2. (IMC 2015, Day 2, Problem 7) *Answer:* the limit equals 1.

Lower bound. For $A > 1$ and $1 \leq x \leq A$ we have $A^{1/x} > 1$, so

$$\frac{1}{A}\int_1^A A^{1/x}\,dx > \frac{1}{A}\int_1^A 1\,dx = 1 - \frac{1}{A}.$$

Upper bound. Fix $\delta > 0$ and $K > 0$. For A large we have $1 < 1+\delta < K\log A < A$. Since $x \mapsto A^{1/x}$ is decreasing, we bound it on each subinterval by its value at the left endpoint:

$$\frac{1}{A}\int_1^A A^{1/x}\,dx = \frac{1}{A}\left(\int_1^{1+\delta} + \int_{1+\delta}^{K\log A} + \int_{K\log A}^{A} \right) < \frac{1}{A}\Big(\delta A + K(\log A) A^{\frac{1}{1+\delta}} + A \cdot A^{\frac{1}{K\log A}} \Big).$$

Now $A^{1/(K\log A)} = e^{1/K}$, so the last expression equals $\delta + KA^{-\delta/(1+\delta)}\log A + e^{1/K}$. Combining the two bounds,

$$1 - \frac{1}{A} < \frac{1}{A}\int_1^A A^{1/x}\,dx < \delta + KA^{-\delta/(1+\delta)}\log A + e^{1/K}.$$

Letting $A \to \infty$ (the middle term $\to 0$) yields

$$1 \leq \liminf_{A\to\infty} \frac{1}{A}\int_1^A A^{1/x}\,dx \leq \limsup_{A\to\infty} \frac{1}{A}\int_1^A A^{1/x}\,dx \leq \delta + e^{1/K}.$$

Finally let $\delta \to 0^+$ and $K \to \infty$; the right side tends to 1. Hence the limit exists and equals 1. $\blacksquare$

Problem 3. (IMC 2017, Day 1, Problem 4)

We use the probabilistic method. Let $d = 1000$, and fix a parameter $p \in (0,1)$. Choose the set S at random, putting each person into S independently with probability p.

For a fixed person v, the event "$v \in S$ and exactly two of v's d friends lie in S" has probability

$$q = p \cdot \binom{d}{2} p^2 (1-p)^{d-2} = \binom{d}{2} p^3 (1-p)^{d-2},$$

since v is included (probability p), and independently exactly 2 of its d friends are chosen while the other $d-2$ are not. Note q does not depend on v.

Now choose $p = \dfrac{3}{d+1}$ (the maximizer of q). Then

$$q = \binom{d}{2} \left(\frac{3}{d+1}\right)^3 \left(\frac{d-2}{d+1}\right)^{d-2} = \frac{27\,d(d-1)}{2(d+1)^3} \left(1 + \frac{3}{d-2}\right)^{-(d-2)}.$$

Using $\left(1+\frac{3}{d-2}\right)^{d-2} < e^3$, we get $\left(1+\frac{3}{d-2}\right)^{-(d-2)} > e^{-3}$, hence

$$q > \frac{27\,d(d-1)}{2(d+1)^3}\,e^{-3}.$$

For $d = 1000$ this gives $q > \dfrac{27\cdot 1000\cdot 999}{2\cdot 1001^3}\,e^{-3} \approx 6.7\cdot 10^{-4} > \dfrac{1}{2017}$.

Let X be the number of persons in S having exactly two friends in S. By linearity of expectation, $\mathbb{E}[X] = nq > \dfrac{n}{2017}$. Since X exceeds its mean for at least one outcome, there is a choice of S with $X > \dfrac{n}{2017}$; in particular at least $n/2017$ persons in S have exactly two friends in S. ■

Problem 4. (IMC 2019, Day 1, Problem 3)

Set

$$g(x) = xf(x) - \frac{x^2}{2}.$$

Then g is twice differentiable and

$$g''(x) = \big(xf(x)\big)'' - 1 = \big(2f'(x) + xf''(x)\big) - 1 \geq 0,$$

so g is convex on $(-1,1)$. A convex function lies above each of its tangent lines; taking the tangent at 0, with $g(0) = 0$ and $g'(0) = a$, gives $g(x) \geq ax$ for all x. Therefore

$$\int_{-1}^{1} xf(x)\,dx = \int_{-1}^{1}\left(g(x) + \frac{x^2}{2}\right)dx \geq \int_{-1}^{1}\left(ax + \frac{x^2}{2}\right)dx = 0 + \frac{1}{3} = \frac{1}{3},$$

since $\int_{-1}^{1} ax\,dx = 0$. ■

Problem 5. (IMC 2015, Day 2, Problem 10)

Let $M = \max_{0\leq x\leq 1}|p(x)|$. For each positive integer k put

$$J_k = \int_0^1 p(x)^{2k}\,dx.$$

Since $p(x)^{2k} \geq 0$ and is not identically 0, we have $0 < J_k < M^{2k}$. Moreover $p(x)^{2k} = \sum_{i=0}^{2kn} c_i x^i$ has integer coefficients, so

$$J_k = \sum_{i=0}^{2kn} \frac{c_i}{i+1}$$

is a positive rational whose denominator divides $\operatorname{lcm}(1,2,\ldots,2kn+1)$. Being positive, it is therefore at least the reciprocal of that number:

$$J_k \geq \frac{1}{\operatorname{lcm}(1,2,\ldots,2kn+1)}.$$

By the prime number theorem, $\log\operatorname{lcm}(1,2,\ldots,N) \sim N$ as $N \to \infty$. Hence for every $\varepsilon > 0$ and all large k,

$$\operatorname{lcm}(1,2,\ldots,2kn+1) < e^{(1+\varepsilon)(2kn+1)},$$

so

$$M^{2k} > J_k \geq \frac{1}{e^{(1+\varepsilon)(2kn+1)}}, \qquad \text{i.e.} \qquad M > \frac{1}{e^{(1+\varepsilon)\left(n+\frac{1}{2k}\right)}}.$$

Letting $k \to \infty$ and then $\varepsilon \to 0^+$ gives $M \geq \frac{1}{e^n}$. Since e is transcendental, equality $M = e^{-n}$ is impossible (it would force a nontrivial algebraic relation for e), so $M > \frac{1}{e^n}$. ■